

CHAPTER 17

yield curve fitting

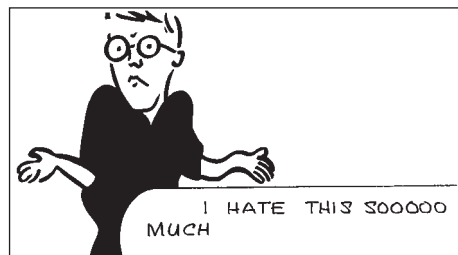


The aim of this Chapter...

... is to show how the simple models can be forced to give a theoretical yield curve which is exactly the same as the market's yield curve. But we also see why this is misleading and dangerous.

In this Chapter...

- how to choose time-dependent parameters in one-factor models so that today's yield curve is an output of the model
- the advantages and disadvantages of yield curve fitting



17.1 INTRODUCTION

One-factor models for the spot rate build up an entire yield curve from a knowledge of the spot rate and the parameters in the model. In using a one-factor model we have to decide how to choose the parameters and whether to believe the output of the model. If we choose parameters using historical time series data then one of the outputs of the model will be a theoretical yield curve. Unless we are very, very lucky this theoretical curve will not be the same as the market yield curve. Which do we believe? Do we believe the theoretical yield curve or do we believe the prices trading in the market? You have to be very brave to ignore the market prices for such liquid instruments as bonds and swaps. Even if you are pricing very complex products you must still hedge with simpler, more liquid, traded contracts for which you would like to get the price right.

Because of this need to correctly price liquid instruments, the idea of **yield curve fitting** or **calibration** has become popular. When one-factor models are used in practice they are almost always fitted. This means that one or more of the parameters in the model is allowed to depend on time. This functional dependence on time is then carefully chosen to make an output of the model, the price of zero-coupon bonds, exactly match the market prices for these instruments. Yield curve fitting is the subject of this chapter.

17.2 HO & LEE

The Ho & Lee spot interest rate model is the simplest that can be used to fit the yield curve. It will be useful to examine this model in detail to see one way in which fitting is done in practice.

In the Ho & Lee model the process for the risk-neutral spot rate is

$$dr = \eta(t)dt + c dX.$$

The standard deviation of the spot rate process, c , is constant, the drift rate η is time dependent.

In this model the solution of the bond pricing equation for a zero-coupon bond is simply

$$Z(r, t; T) = e^{A(t; T) - r(T-t)},$$

where

$$A(t; T) = - \int_t^T \eta(s)(T-s)ds + \frac{1}{6}c^2(T-t)^3.$$

If we know $\eta(t)$ then the above gives us the theoretical value of zero-coupon bonds of all maturities. Now turn this relationship around and ask the question: 'What functional form must we choose for $\eta(t)$ to make the theoretical value of the discount rates for all maturities equal to the market values?' Call this special choice for η , $\eta^*(t)$. The yield curve is to be fitted today, $t = t^*$, when the spot interest rate is r^* and the discount factors in the market are $Z_M(t^*; T)$. To match the market and theoretical bond prices, we must solve

$$Z_M(t^*; T) = e^{A(t^*; T) - r^*(T-t^*)}.$$

Taking logarithms of this and rearranging slightly we get

$$\int_{t^*}^T \eta^*(s)(T-s)ds = -\log(Z_M(t^*; T)) - r^*(T-t^*) + \frac{1}{6}c^2(T-t^*)^3. \quad (17.1)$$

Observe that I am carrying around in the notation today's date t^* . This is a constant but I want to emphasize that we are doing the calibration to *today's* yield curve. If we calibrate again tomorrow, the market yield curve will have changed.

Differentiate (17.1) twice with respect to T to get

$$\eta^*(t) = c^2(t-t^*) - \frac{\partial^2}{\partial t^2} \log(Z_M(t^*; t)).$$

With this choice for the time-dependent parameter $\eta(t)$ the theoretical and actual market prices of zero-coupon bonds are the same. It also follows that

$$A(t; T) = \log \left(\frac{Z_M(t^*; T)}{Z_M(t^*; t)} \right) - (T-t) \frac{\partial}{\partial t} \log(Z_M(t^*; t)) - \frac{1}{2}c^2(t-t^*)(T-t)^2.$$

17.3 THE EXTENDED VASICEK MODEL OF HULL & WHITE

The Ho & Lee model isn't the only one that can be calibrated, it's just the easiest. Most one-factor models have the potential for fitting, but the more tractable the model the easier the fitting. If the model is not at all tractable, having no nice explicit zero-coupon bond price formula, then we can always resort to numerical methods.

The next easiest model to fit is the Vasicek model. The Vasicek model has the following stochastic differential equation for the risk-neutral spot rate

$$dr = (\eta - \gamma r)dt + c dX.$$

Hull & White extend this to include a time-dependent parameter

$$dr = (\eta(t) - \gamma r)dt + c dX.$$

Assuming that γ and c have been estimated statistically, say, we choose $\eta = \eta^*(t)$ at time t^* so that our theoretical and the market prices of bonds coincide.

Under this risk-neutral process the value of a zero-coupon bonds

$$Z(r, t; T) = e^{A(t; T) - rB(t; T)},$$

where

$$A(t; T) = -\int_t^T \eta^*(s)B(s; T)ds + \frac{c^2}{2\gamma^2} \left(T-t + \frac{2}{\gamma}e^{-\gamma(T-t)} - \frac{1}{2\gamma}e^{-2\gamma(T-t)} - \frac{3}{2\gamma} \right).$$

and

$$B(t; T) = \frac{1}{\gamma} (1 - e^{-\gamma(T-t)}).$$

To fit the yield curve at time t^* we must make $\eta^*(t)$ satisfy

$$\begin{aligned} A(t^*; T) &= - \int_{t^*}^T \eta^*(s) B(s; T) ds + \frac{c^2}{2\gamma^2} \left(T - t^* + \frac{2}{\gamma} e^{-\gamma(T-t^*)} - \frac{1}{2\gamma} e^{-2\gamma(T-t^*)} - \frac{3}{2\gamma} \right) \\ &= \log(Z_M(t^*; T)) + r^* B(t^*, T). \end{aligned} \quad (17.2)$$

This is an integral equation for $\eta^*(t)$ if we are given all of the other parameters and functions, such as the market prices of bonds, $Z_M(t^*; T)$.

Although (17.2) may be solved by Laplace transform methods, it is particularly easy to solve by differentiating the equation twice with respect to T . This gives

$$\eta^*(t) = - \frac{\partial^2}{\partial t^2} \log(Z_M(t^*; t)) - \gamma \frac{\partial}{\partial t} \log(Z_M(t^*; t)) + \frac{c^2}{2\gamma} \left(1 - e^{-2\gamma(t-t^*)} \right). \quad (17.3)$$

From this expression we can now find the function $A(t; T)$,

$$\begin{aligned} A(t; T) &= \log \left(\frac{Z_M(t^*; T)}{Z_M(t^*; t)} \right) - B(t; T) \frac{\partial}{\partial t} \log(Z_M(t^*; t)) \\ &\quad - \frac{c^2}{4\gamma^3} \left(e^{-\gamma(T-t^*)} - e^{-\gamma(t-t^*)} \right)^2 \left(e^{2\gamma(t-t^*)} - 1 \right). \end{aligned}$$

17.4 YIELD-CURVE FITTING: FOR AND AGAINST

17.4.1 For

The building blocks of the bond pricing equation are delta hedging and no arbitrage. If we are to use a one-factor model correctly then we must abide by the delta-hedging assumptions. We must buy and sell instruments to remain delta neutral. The buying and selling of instruments must be done at the market prices. We *cannot* buy and sell at a theoretical price. But we are not modeling the bond prices directly; we model the spot rate and bond prices are then derivatives of the spot rate. This means that there is a real likelihood that our output bond prices will differ markedly from the market prices. This is useless if we are to hedge with these bonds. The model thus collapses and cannot be used for pricing other instruments, unless we can find a way to generate the correct prices for our hedging instruments from the model; this is yield curve fitting.

Once we have fitted the prices of traded products we then dynamically or statically hedge with these products. The idea being that even if the model is wrong so that we lose money on the contract we are pricing then we should make that money back on the hedging instruments.

17.4.2 Against

If the market prices of simple bonds were correctly given by a model, such as Ho & Lee or Hull & White, fitted at time t^* then, when we come back a week later, $t^* + \text{one week}$, say, to refit the function $\eta^*(t)$, we would find that this function *had not changed* in the

meantime. This *never* happens in practice. We find that the function η^* has changed out of all recognition. What does this mean? Clearly the model is wrong.¹

By simply looking for a Taylor series solution of the bond pricing equation for short times to expiry, we can relate the value of the risk-adjusted drift rate at the short end to the slope and curvature of the market yield curve. This is done as follows. Look for a solution of (16.4) of the form

$$Z(r, t; T) \sim 1 + a(r)(T - t) + b(r)(T - t)^2 + c(r)(T - t)^3 + \dots$$

Substitute this into the bond pricing equation:

$$\begin{aligned} & -a - 2b(T - t) - 3c(T - t)^2 + \frac{1}{2} \left(w^2 - 2(T - t)w \frac{\partial w}{\partial t} \right) \left((T - t) \frac{d^2 a}{dr^2} + (T - t)^2 \frac{d^2 b}{dr^2} \right) \\ & + \left((u - \lambda w) - (T - t) \frac{\partial(u - \lambda w)}{\partial t} \right) (T - t) \left(\frac{da}{dr} + (T - t)^2 \frac{db}{dr} \right) \\ & - r \left(1 + a(T - t) + c(T - t)^2 \right) + \dots = 0. \end{aligned}$$

Note how I have expanded the drift and volatility terms about $t = T$, in the above these are evaluated at r and T . By equating powers of $(T - t)$ we find that

$$a(r) = -r, \quad b(r) = \frac{1}{2}r^2 - \frac{1}{2}(u - \lambda w)$$

and

$$\begin{aligned} c(r) = & \frac{1}{12}w^2 \frac{\partial^2}{\partial r^2} (r^2 - r(u - \lambda w)) - \frac{1}{6}(u - \lambda w) \frac{\partial}{\partial r} (r^2 - r(u - \lambda w)) \\ & - \frac{1}{3} \frac{\partial}{\partial t} (u - \lambda w) + \frac{1}{6}r^2 (r - (u - \lambda w)). \end{aligned}$$

In all of these $u - \lambda w$ and w are evaluated at r and T .

From the Taylor series expression for Z we find that the yield to maturity is given by

$$-\frac{\log(Z(r, t; T))}{T - t} \sim -a + \left(\frac{1}{2}a^2 - b \right) (T - t) + \left(ab - c - \frac{1}{3}a^3 \right) (T - t)^2 + \dots$$

for short times to maturity.

The yield curve takes the value $-a(r) = r$ at maturity, obviously. The slope of the yield curve is

$$\frac{1}{2}a^2 - b = \frac{1}{2}(u - \lambda w),$$

i.e. one half of the risk-neutral drift. The curvature of the yield curve at the short end is proportional to

$$ab - c - \frac{1}{3}a^3,$$

¹ This doesn't mean that it isn't useful, or profitable. This is a much more subtle point.

which contains a term that is the derivative of the risk-neutral drift with respect to time via c . Let me stress the key points of this analysis. The slope of the yield curve at the short end depends on the risk-neutral drift, and vice versa. The curvature of the yield curve at the short end depends on the time derivative of the risk-neutral drift, and vice versa.

If we choose time-dependent parameters within the risk-adjusted drift rate such that the market prices are fitted at time t^* then we have

$$Z(r^*, t^*; T) = Z_M(t^*; T)$$

which is one equation for the time-dependent parameters.

Thus, for Ho & Lee, for example, the value of the function $\eta^*(t)$ at the short end, $t = t^*$, depends on the slope of the market yield curve. Moreover, the slope of $\eta^*(t)$ depends on the *curvature* of the yield curve at the short end. Results such as these are typical for all fitted models. These, seemingly harmless, results are actually quite profound.

It is common for the slope of the yield curve to be quite large and positive, the difference between very short and not quite so short rates is large. But then for longer maturities typically the yield curve flattens out. This means that the yield curve has a large negative curvature. If one performs the fitting procedure as outlined here for the Ho & Lee or extended Vasicek models, one typically finds the following:

- The value of $\eta^*(t)$ at $t = t^*$ is very large. This is because the yield curve slope at the short end is often large.
- The slope of $\eta^*(t)$ at $t = t^*$ is large and negative. This is because the curvature of the yield curve is often large and negative.

A typical plot of $\eta^*(t)$ versus t is shown in Figure 17.1. This shows the high value for the fitted function and the large negative slope.² So far, so good. Maybe this is correct, maybe this is really what the fitted parameter should look like. But what happens when we come back in a few months to look at how our fitted parameter is doing? If the model is correct then we would find that the fitted curve looked like the bold part of the curve in the figure. The previous data should have just dropped off the end, the rest of the curve should remain unchanged. We would then see a corresponding dramatic flattening of the yield curve. Does this in fact happen? No. The situation looks more like that in Figure 17.2, which is really just a translation of the curve in time. Again we see the high value at the short end, the large negative slope and the oscillations. The recalibrated function in Figure 17.2 looks nothing like the bold line in Figure 17.1. This is because the yield curve has not changed that much in the meantime. It still has the high slope and curvature. In fact, we don't even have to wait for a few months for the deviation to be significant, it becomes apparent in weeks or even days.

We can conclude from this that yield curve fitting is an inconsistent and dangerous business. The results presented here are by no means restricted to the models I have named; *no* one-factor model will capture the high slope and curvature that is usual for yield curves; they 'may' give reasonable results when the yield curve is fairly flat.

² The strange oscillation of the function η^* beyond the short end is usually little more than numerical errors.

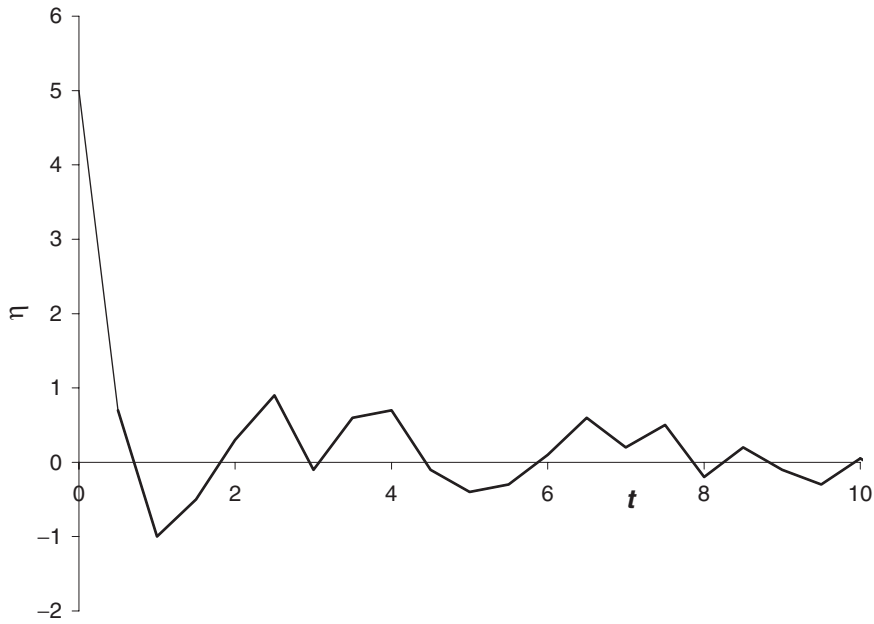


Figure 17.1 Typical fitted function $\eta^*(t)$.

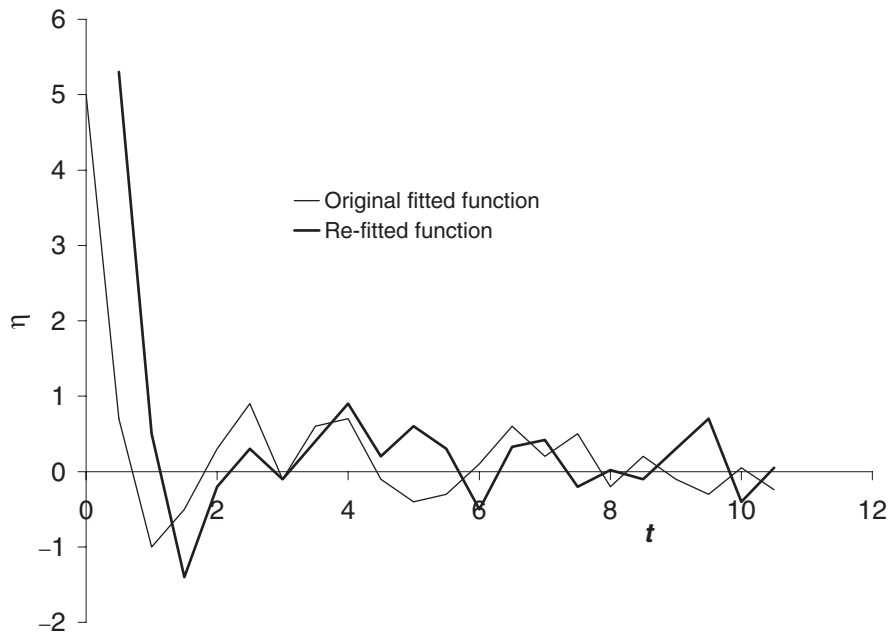


Figure 17.2 Typical re-fitted function $\eta^*(t)$, a short time later.

17.5 OTHER MODELS

Other models for the short-term interest rate have been proposed. One of the most popular (but for which there are no explicit solutions) is the **Black, Derman & Toy (BDT)** model where the risk-neutral spot interest rate satisfies

$$d(\log r) = \left(\theta(t) - \frac{\sigma'(t)}{\sigma(t)} \log r \right) dt + \sigma(t) dX.$$

The two functions of time σ and θ allow both zero-coupon bonds and their volatilities to be matched. An even more general model is the **Black & Karasinski** model

$$d(\log r) = (\theta(t) - a(t) \log r) dt + \sigma(t) dX.$$

These models are popular because fitting can be done quite simply by a numerical scheme.

Any criticisms of yield curve fitting in general, of course, apply to these models.

17.6 SUMMARY

I have outlined why the yield curve is fitted, and how it is fitted in some simple models. From a practical perspective it is hard to argue against calibration, you cannot hedge with something if your theoretical price is very different from its traded price. But from a modeling and empirical perspective it is hard to argue in its favor, the data shows how inconsistent the concept is. This is always going to be a problem with one-factor Brownian motion models, unless yield curves suddenly decide not to be so steep. There is not a great deal that can be done theoretically.

On the other hand, people seem to make money using these models and I guess that is the correct test of a model. Unless you are speculating with an interest rate derivative, you will have to delta hedge and therefore have to calibrate. Practitioners go much further than I have shown here, they fit as many market prices and properties as they can. Put in another time-dependent parameter and you can fit interest rate volatilities of different maturities, yet another parameter and you can fit the market prices of caps. By fitting more and more data, are you digging a deeper and deeper grave or are you improving and refining the accuracy of your model?

As an aside, suppose we are not interested in hedging but want to speculate with some fixed-income instruments. It is common knowledge that the yield curve is a poor predictor of real future interest rates. In this case it could be unnecessary or even dangerous to fit the yield curve. In this situation one could 'value' the instrument using the *real* spot rate process. This would give a 'value' for the instrument that was the expected present value of all cashflows under the *real* random walk. To do this one needs a model for the real drift u .

FURTHER READING

- A more sophisticated choice of time-dependent parameters is described by Hull & White (1990a).

- Klugman & Wilmott (1994) consider the fitting of the general affine model.
- Baker (1977) gives details of the numerical solution of integral equations.
- See Black *et al.* (1990) for details of their popular model.
- Rebonato (1996) discusses calibration in depth for many popular models.
- See Derman (2004) for the story behind the BDT model.
- In *PWOQF2* is shown a non-linear interest rate model that completely sidesteps the issue of calibration.

EXERCISES

1. Substitute the fitted function for $A(t; T)$, using the Ho & Lee model, back into the solution of the bond pricing equation for a zero-coupon bond,

$$Z(r, t; T) = e^{A(t; T) - r(T-t)}.$$

What do you notice?

2. Differentiate Equation (17.2) twice to solve for the value of $\eta^*(t)$. What is the value of a zero-coupon bond with a fitted Vasicek model for the interest rate?
3. Use market data for the price of zero-coupon bonds to fit a Ho & Lee model. Refit the model with data for a week later in time. Compare the two curves for $\eta^*(t)$.

Note: The second curve for η^* starts a week after the first curve. They should not be plotted starting at the same point in time.

4. Use market data for zero-coupon bond prices to fit a Vasicek model for the interest rate.

